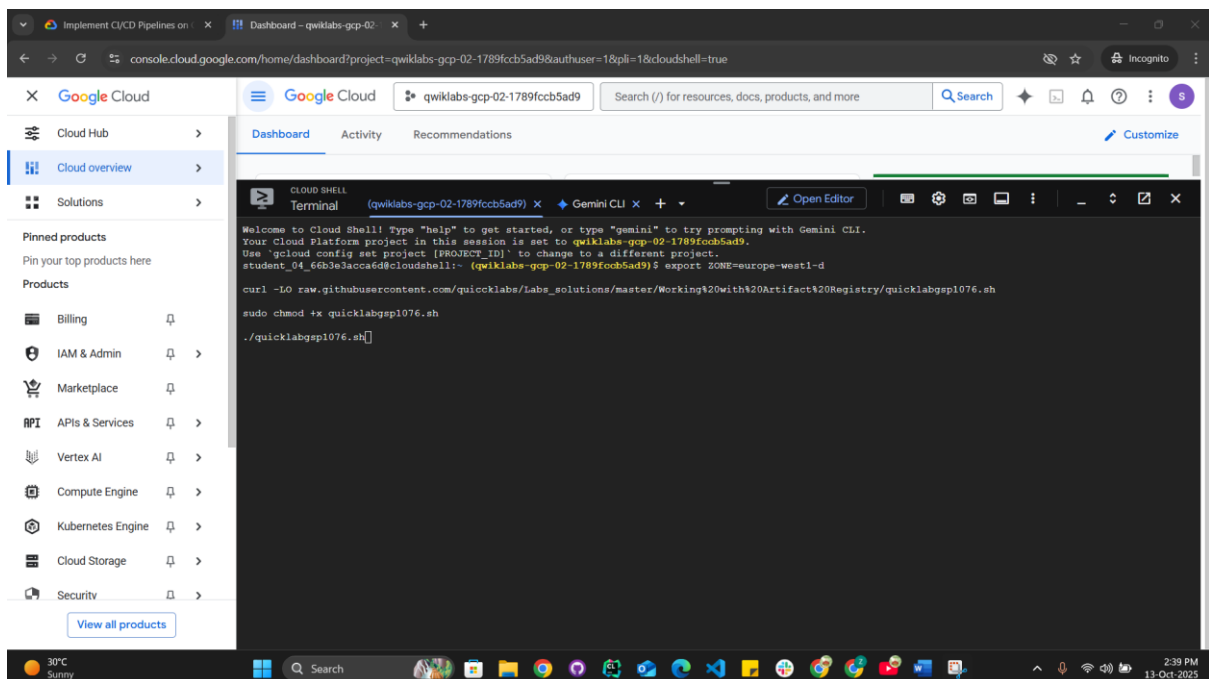
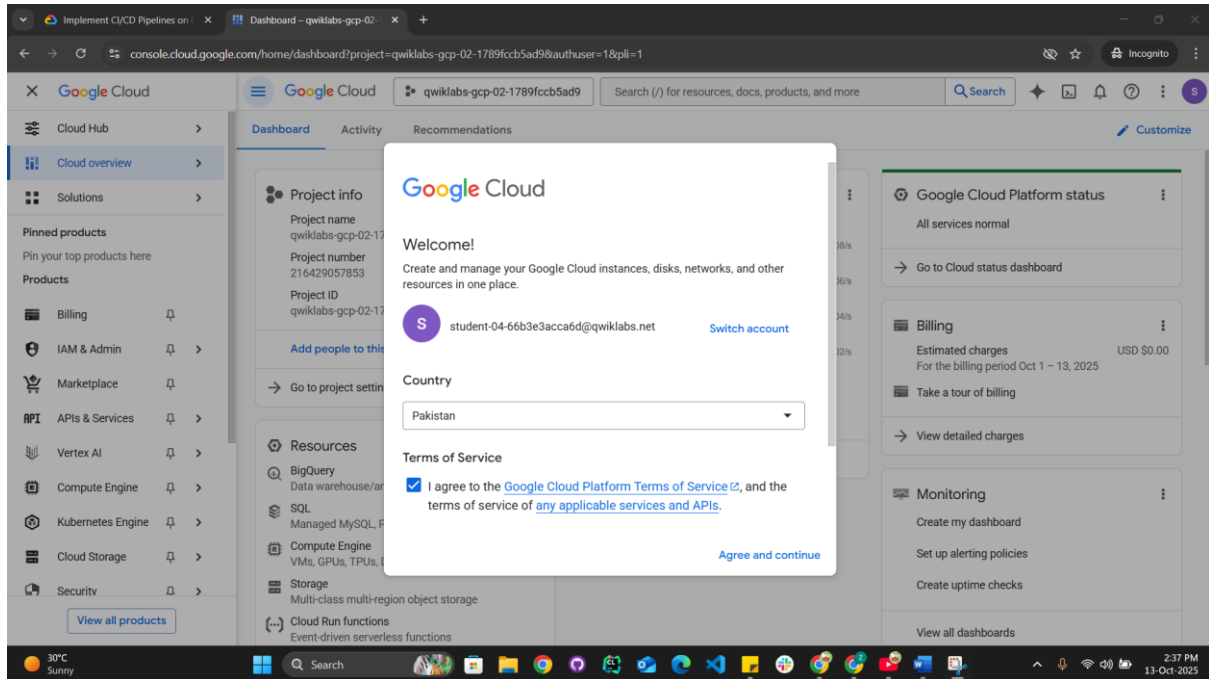
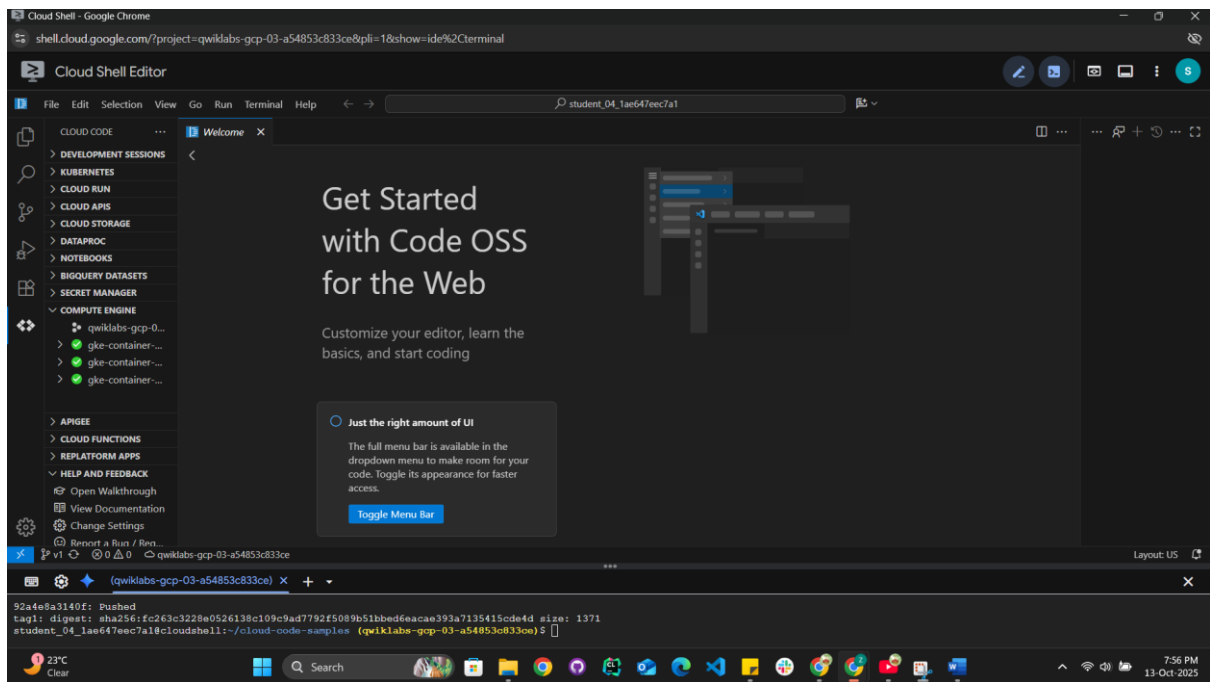
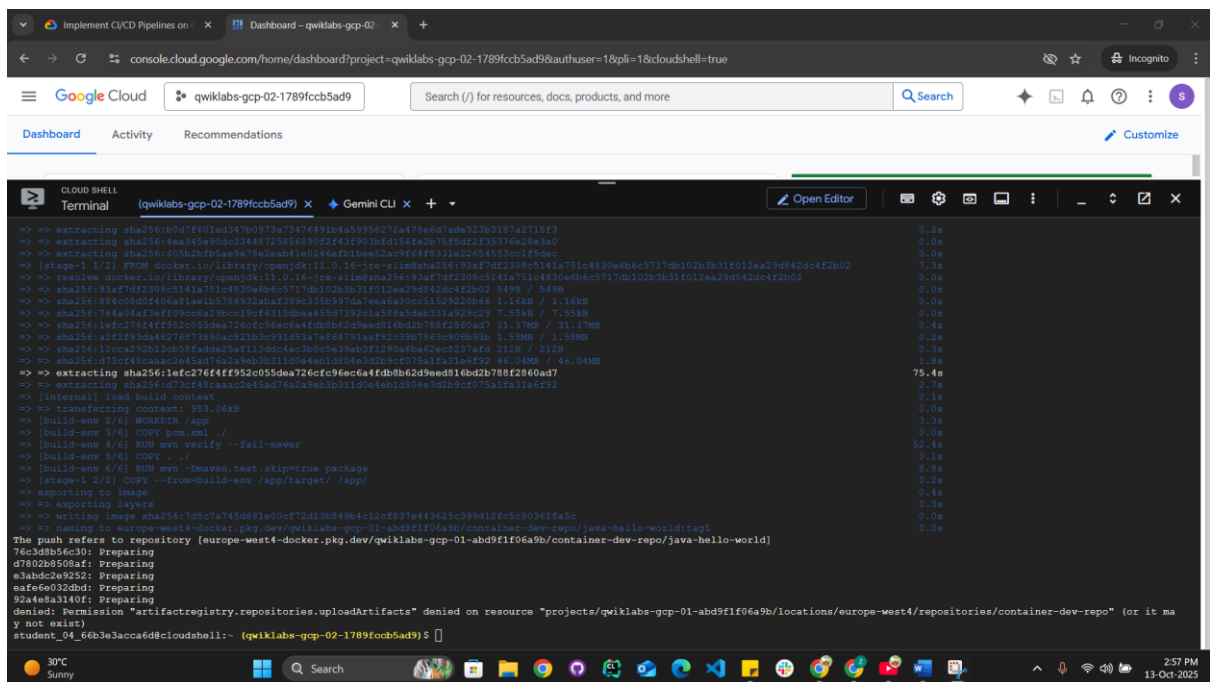
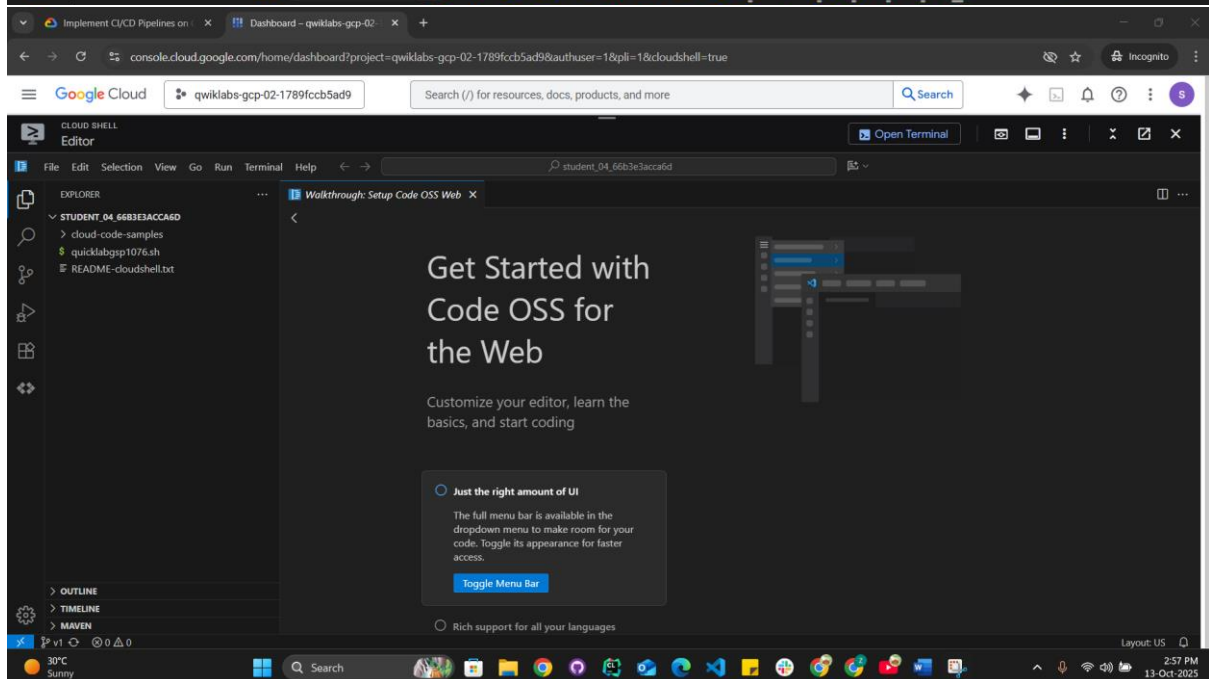
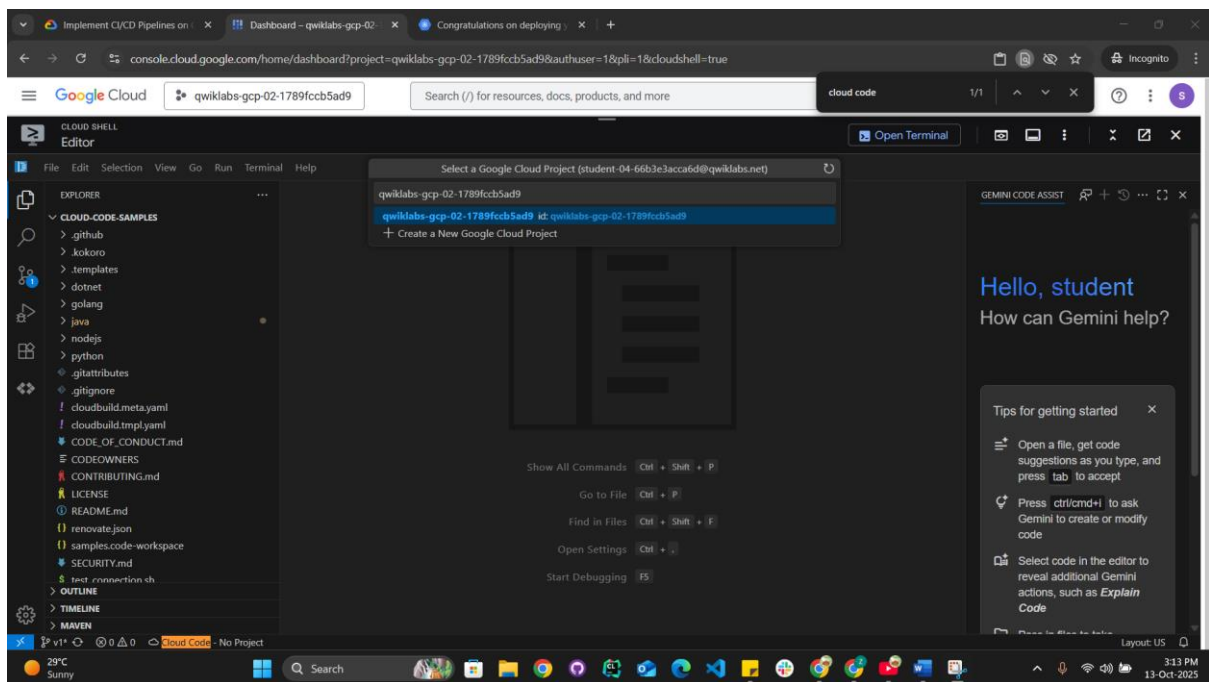
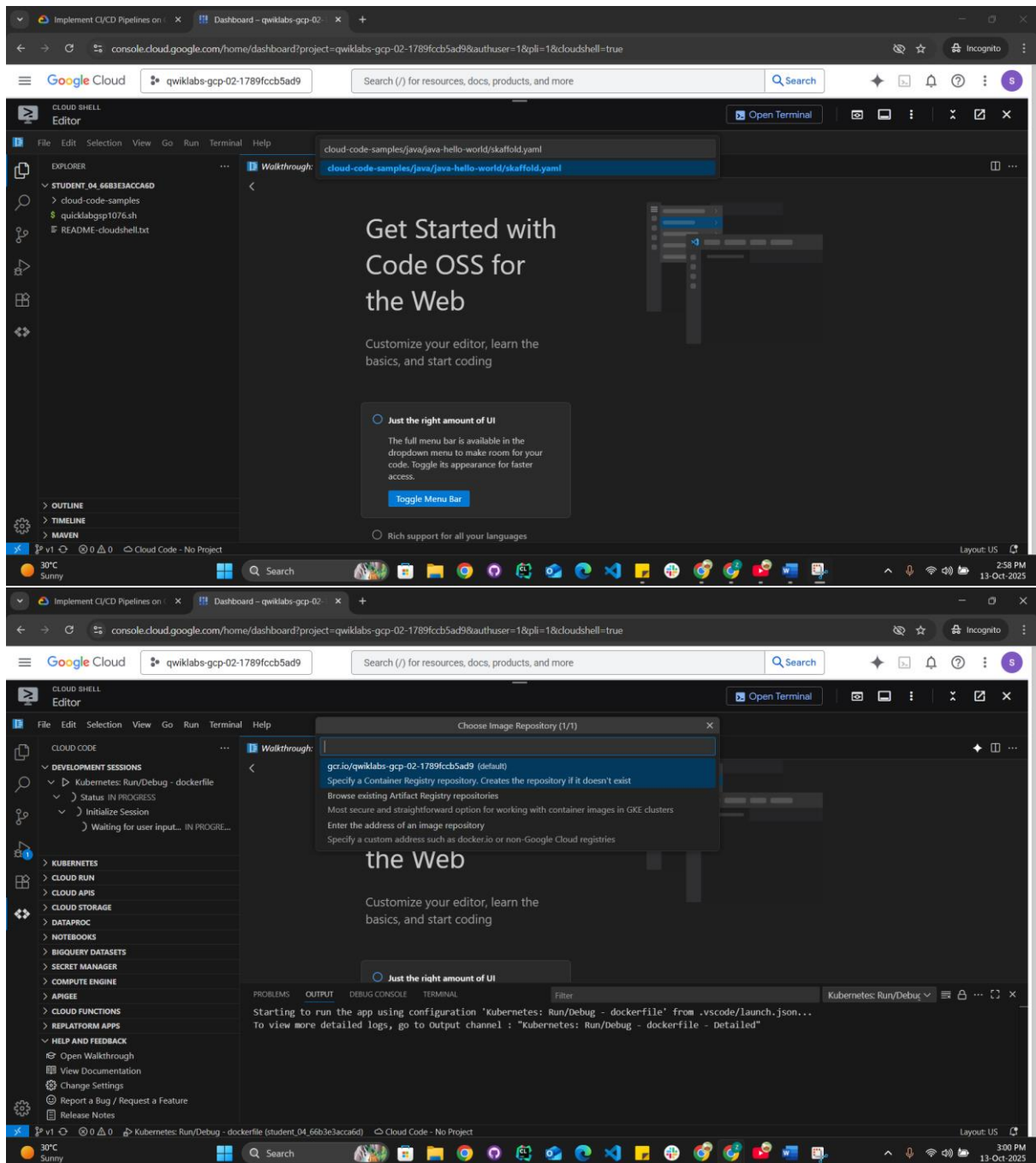


1. Working with Artifact Registry









Implement CI/CD Pipelines on ...Dashboard - qwiklabs-gcp-02Loading...console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-02-1789fcb5ad9&authuser=1&pli=1&cloudshell=true

Google Cloudqwiklabs-gcp-02-1789fcb5ad9Search (/) for resources, docs, products, and moreSearch

CLOUD SHELLEditor

FileEditSelectionViewGoRunTerminalHelpstudent_04_66b3e3acca6dOpen Terminal

CLOUD CODE

DEVELOPMENT SESSIONS

KUBERNETES

CLOUD RUN

CLOUD APIS

CLOUD STORAGE

DATAPROC

NOTEBOOKS

BIGQUERY DATASETS

SECRET MANAGER

COMPUTE ENGINE

APIGEE

CLOUD FUNCTIONS

REPLATFORM APPS

HELP AND FEEDBACK

Open Walkthrough

View Documentation

Change Settings

Report a Bug / Request a Feature

Release Notes

Walkthrough: Setup Code OSS Web

Get Started with Code OSS for the Web

Customize your editor, learn the basics, and start coding

Just the right amount of UI

PROBLEMSOUTPUTDEBUG CONSOLETERMINALFilterKubernetes: Run/Debug

Resource deployment/java-hello-world status completed successfully
Status check succeeded
*****URLS*****
Deploy completed
Forwarded URL from service java-hello-world-external: http://localhost:4503
Update succeeded

v1000Kubernetes: Run/Debug - dockerfile (student_04_66b3e3acca6d)Cloud Code - No Project

Deployment SuccessfulLayout: US

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Search

3:03 PM13-Oct-2025

Implement CI/CD Pipelines on ...Dashboard - qwiklabs-gcp-02Congratulations on deploying ...4503-cs-111092e2-f7f8-49bf-887b-4ebd9fb0d0fb.qf-asia-southeast1-wlry.cloudshell.dev

It's running!

Congratulations, you successfully deployed a Kubernetes application with Cloud Code!

What's next?

Learn more about working with Kubernetes in Cloud Code for [VSCode](#) or [IntelliJ](#)

Get started with [Google Cloud APIs](#) in Cloud Code

Watch how Cloud Code can optimize your [GKE Developer Workflow](#)

Explore more options with

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Implement CI/CD Pipelines on > Dashboard - qwiklabs-gcp-02 > Congratulations on deploying > +

console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-02-1789fcb5ad9&authuser=1&pli=1&cloudshell=true

Google Cloud qwiklabs-gcp-02-1789fcb5ad9 Search (/) for resources, docs, products, and more

CLOUD SHELL Editor

File Edit Selection View Go Run Terminal Help

student_04_66b3e3acca6d

Walkthrough: Setup Code OSS Web Hello

cloud-code-samples > java > java-hello-world > src > main > java > cloudcode > helloworld > web > HelloWorldController.java

```
1 package cloudcode.helloworld.web;
2
3
4 import org.springframework.stereotype.Controller;
5 import org.springframework.ui.Model;
6 import org.springframework.web.bind.annotation.GetMapping;
7
8 /**
9  * Defines a controller to handle HTTP requests.
10 */
11 @Controller
12 public final class HelloWorldController {
13
14     /**
15      * Create an endpoint for the landing page
16      * @return the index view template with a simple message
17      */
18     @GetMapping("/")
19     public String helloworld(Model model) {
20         String message = "It's updated!";
21         model.addAttribute("message", message);
22         return "index";
23     }
24 }
25
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL Filter

Kubernetes: Run/Debug

Deploy completed

Forwarded URL from service java-hello-world-external: <http://localhost:4503>

Update succeeded

Watching for changes...

To disable watch mode for subsequent runs, set watch to false in your launch configuration /home/student_04_66b3e3acca6d/.vscode/launch.json and relaunch the application.

Update initiated

Do you want to install the recommended 'Extension Pack for Java' extension from vsjava for the Java language? [Install](#) [Show Recommendations](#)

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Implement CI/CD Pipelines on... Dashboard - qwiklabs-gcp-02... Congratulations on deploying... 4503-cs-11109262-f7f8-49bf-887b-4ebd9b0d0fb.q1-asia-southeast1-wly.cloudshell.dev

It's updated!

Congratulations, you successfully deployed a Kubernetes application with Cloud Code!

What's next?

Learn more about working with Kubernetes in Cloud Code for [VSCode](#) or [IntelliJ](#)

Get started with [Google Cloud APIs](#) in Cloud Code

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console.cloud.google.com/home/dashboard?project=qwiklabs-gcp-02-1789fccb5ad9&authuser=1&pli=1&cloudshell=true

Google Cloud qwiklabs-gcp-02-1789fccb5ad9 Search (/) for resources, docs, products, and more

CLOUD SHELL Terminal (quiklabs-gcp-02-1789fccb5ad9) Gemini CLI + Open Editor

```
>> extracting sha256:80b2b3f3a9e78e1e04e4f1b2e82c0f44f8331e225848530c1f5d8c 0.0s
>> [stage-1 1/2] RUN docker.io/library/openjdk:11.0.14-jre-slim&sha256:93af7df2200c5141a751c4830e4bdc5717db102b3b31f012ea294842dc4f2b02 7.3s
>> resolve docker.io/library/openjdk:11.0.14-jre-slim&sha256:93af7df2200c5141a751c4830e4bdc5717db102b3b31f012ea294842dc4f2b02 0.0s
>> sha256:93af7df2200c5141a751c4830e4bdc5717db102b3b31f012ea294842dc4f2b02 549B / 549B 0.0s
>> sha256:93af7df2200c5141a751c4830e4bdc5717db102b3b31f012ea294842dc4f2b02 1.1KB / 1.1KB 0.0s
>> sha256:76a044f3eef0b0ca29b0c158d0a485d392c1a33845d0031a928c9 7.55MB 0.0s
>> sha256:1efc2764ff952c055dea726fc96ec64fdb8b61d9eed16bd2b788f22860ad7 31.37MB / 31.37MB 0.4s
>> sha256:a22f95da48276873890ac821b3c991d51a7a864791aaf92c39b7863c90bb93b 1.58MB / 1.58MB 0.2s
>> sha256:12c0e49a1b12b284d0a5a1110b4f6c0c343ad31f29a040e0c227e48 212B / 212B 0.0s
>> sha256:d73cf40caac0e45d1f0ca9eb3b311d0e4eb1d80e3d2b5cf075a1fa31e6f92 46.04MB / 46.04MB 1.8s
>> extracting sha256:1efc2764ff952c055dea726fc96ec64fdb8b61d9eed16bd2b788f22860ad7 75.4s
>> extracting sha256:d73cf40caac0e45d1f0ca9eb3b311d0e4eb1d80e3d2b5cf075a1fa31e6f92 2.7s
>> [itwenal] /usr build context 2.1s
>> transferring context: 851.0KB 0.0s
>> [build-env 2/6] WORKDIR /app 3.3s
>> [build-env 3/6] COPY pom.xml ./ 0.0s
>> [build-env 4/6] RUN mvn verify --fail-never 52.4s
>> [build-env 5/6] COPY ./ 0.1s
>> [build-env 6/6] RUN mvn -Dmaven.test.skip=true package 8.8s
>> [stage-1 2/2] COPY --from=build-env /app/target/ /app/ 0.2s
>> exporting to image 0.4s
>> exporting layers 0.3s
>> writing image sha256:7d8c7a745d4e1e00c72d12b849b4c12c837e443625c399d12fcb90361fa3c 0.0s
>> pushing to europe-west4-docker.pkg.dev/quiklabs-gcp-01-abd9f1f06a9b/container-dev-repo/java-hello-world:tag1 0.0s
The push refers to repository (europe-west4-docker.pkg.dev/quiklabs-gcp-01-abd9f1f06a9b/container-dev-repo/java-hello-world:tag1)
76c3d9b56c30: Preparing
d7802b508af: Preparing
e1abd2e9252: Preparing
eafe60328bd: Preparing
92a4e8a3140f: Preparing
denied: Permission "artifactregistry.repositories.uploadArtifacts" denied on resource "projects/quiklabs-gcp-01-abd9f1f06a9b/locations/europe-west4/repositories/container-dev-repo" (or it may not exist)
student_04_6b3e3acca6@cloudshell:~ (quiklabs-gcp-02-1789fccb5ad9) $ gcloud artifacts repositories create container-dev-java-repo \
--repository-format=maven \
--location=europe-west1 \
--description="Java package repository for Container Dev Workshop"
Create request issued for: [container-dev-java-repo]
Waiting for operation [projects/quiklabs-gcp-02-1789fccb5ad9/locations/europe-west1/operations/4242fa00-49c6-461d-9370-062aead5611] to complete....done.
Created repository [container-dev-java-repo].
student_04_6b3e3acca6@cloudshell:~ (quiklabs-gcp-02-1789fccb5ad9) $
```

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Software Development & Cloud | Implement CI/CD Pipelines on Google Cloud

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10 Implement CI/CD Pipelines on Google Cloud

Course - 1 Complete hour 15 minutes

[Course overview](#)

Implement CI/CD Pipelines on Google Cloud

- Working with Artifact Registry
- Google Kubernetes Engine Pipeline using Cloud Build
- Continuous Delivery with Google Cloud Deploy

End Lab 00:30:29

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

[Open Console](#)

Project ID: qwiklabs-gcp-03-a54853cl

Username: student-04-1ae647eec7a1l

Password: iLv3f3sXh9IS

3. In the Cloud console go to **Artifact Registry > Repositories** and notice your newly created Maven repository named `container-dev-java-repo`, if you click on it you can see that it's empty at the moment.

Click **Check my progress** to verify the objective.

Create a Java package repository

[Check my progress](#)

Set up authentication to Artifact Repository

- Use the following command to update the well-known location for Application Default Credentials (ADC) with your user account credentials so that the Artifact Registry credential helper can authenticate using them when connecting with

[Check complete. Points earned: 20. Message: Assessment Completed!](#)

[Next >](#)

Lab instructions and tasks 100/100

GSP1076

Overview

Objectives

Setup and requirement

Task 1. Prepare the lab environment

Task 2. Working with container images

Task 3. Integration with Cloud Code

Task 4. Working with language packages

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10 Implement CI/CD Pipelines on Google Cloud

Course - 1 Complete hour 15 minutes

[Course overview](#)

Implement CI/CD Pipelines on Google Cloud

- Working with Artifact Registry
- Google Kubernetes Engine Pipeline using Cloud Build
- Continuous Delivery with Google Cloud Deploy

End Lab 00:30:04

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

[Open Console](#)

Project ID: qwiklabs-gcp-03-a54853cl

Username: student-04-1ae647eec7a1l

Password: iLv3f3sXh9IS

3. In the Cloud console go to **Artifact Registry > Repositories** and notice your newly created Maven repository named `container-dev-java-repo`, if you click on it you can see that it's empty at the moment.

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Create a Java package repository

[Check my progress](#)

Set up authentication to Artifact Repository

- Use the following command to update the well-known location for Application Default Credentials (ADC) with your user account credentials so that the Artifact Registry credential helper can authenticate using them when connecting with

Checkpoints

- Enable Google Services and create container cluster [Check my progress](#) 40 / 40
- Working with container images [Check my progress](#) 20 / 20
- Integration with Cloud Code [Check my progress](#) 20 / 20

[Next >](#)

Lab instructions and tasks

GSP1076

Overview

Objectives

Setup and requirement

Task 1. Prepare the lab environment

Task 2. Working with container images

Task 3. Integration with Cloud Code

Task 4. Working with language packages

Software Development & Con...Implement CI/CD Pipelines on...cloudskillsboost.google/paths/76/course_templates/691/labs/594611

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image1/29Z

Apply your skills in Google Cloud consoleGet started

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10Implement CI/CD Pipelines on Google Cloud

Course - 1Completehour 15 minutes

Course overview

Implement CI/CD Pipelines on Google Cloud

Working with Artifact Registry

Google Kubernetes Engine Pipeline using Cloud Build

Continuous Delivery with Google Cloud Deploy

End Lab00:29:54

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Console

Project IDqw1klabs-gcp-03-a54853cl

Usernamestudent-04-1ae647eec7a1l

PasswordiLv93fsXh9IS

Previous

3. In the Cloud console go to **Artifact Registry**. On the left sidebar, click **Repositories** and notice your newly created Maven repository named **cd**. On it you can see that it's empty at the moment.

Click **Check my progress** to verify the objective.

Create a Java package repository

Check my progress

Set up authentication to Artifact Repository

Use the following command to update the well-known location for Application Default Credentials (ADC) with your user account credentials so that the Artifact Registry credential helper can authenticate using them when connecting with

Working with container images

Check my progress20 / 20

Integration with Cloud Code

Check my progress20 / 20

Create a Java package repository

Check my progress20 / 20

environment

Task 2. Working with container images

Task 3. Integration with Cloud Code

Task 4. Working with language packages

Next

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